

Theory Behind RSA Cryptography

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Introduction to RSA

RSA cryptography is an asymmetric encryption algorithm based on **Number Theory**, specifically **modular arithmetic**.

The Core Idea

The security of RSA relies on the practical difficulty of factoring the product of two large prime numbers (the factoring problem).

Step 1: Key Generation (The Setup)

To set up the system, follow these steps:

- 1 Select two large distinct prime numbers, p and q .
- 2 Compute $n = pq$.
- 3 Calculate the totient $\phi(n) = (p - 1)(q - 1)$.
- 4 Choose an integer e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$.
- 5 Determine d as the modular multiplicative inverse of e :

$$ed \equiv 1 \pmod{\phi(n)}$$

Step 2: Encryption and Decryption

Let M be the plaintext message (represented as a number).

Encryption

The ciphertext C is computed using the public key (e, n) :

$$C \equiv M^e \pmod{n}$$

Decryption

The original message M is recovered using the private key (d, n) :

$$M \equiv C^d \pmod{n}$$

To understand why this works, we need two key concepts:

1. Fermat's Little Theorem / Euler's Theorem

If $\gcd(a, n) = 1$, then:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

2. Modular Inverse

An inverse d exists for $e \pmod{\phi(n)}$ if and only if $\gcd(e, \phi(n)) = 1$. This is why we choose e to be coprime to $\phi(n)$.

The Proof of Correctness

Starting with the decryption formula:

$$\begin{aligned} C^d &\equiv (M^e)^d \pmod{n} \\ &\equiv M^{ed} \pmod{n} \end{aligned}$$

Since $ed \equiv 1 \pmod{\phi(n)}$, we can write $ed = k\phi(n) + 1$ for some integer k :

$$\begin{aligned} M^{k\phi(n)+1} &\equiv (M^{\phi(n)})^k \cdot M^1 \pmod{n} \\ &\equiv (1)^k \cdot M \pmod{n} \\ &\equiv M \pmod{n} \end{aligned}$$

The original message is successfully recovered!